CMMI V3.0

Model Reference & Study Summary

A structured summary of the CMMI Model V3.0 (ISACA, 2023) — model architecture, the four Categories and eleven Capability Areas, the eight Domains, all 31 Practice Areas with their intent, value and practices by level, practice group levels, maturity and capability levels, adoption guidance, process habit and persistence, high maturity, glossary and abbreviations.

Contents

How to read this document. Sections 1–6 cover the model's structure (Part One of the source). Sections 7–9 correspond to Parts Two, Three, and Four. Section 10 is the working reference: one entry per Practice Area, giving its intent, business value, and every practice organised by practice group level.

1. About CMMI

Capability Maturity Model Integration (CMMI) is an integrated set of best practices that enables organizations to improve the performance of their key business processes. It was developed by product teams drawn from industry and the CMMI Institute, now part of ISACA. At its heart the CMMI provides a roadmap for **building, improving, and sustaining capability** — guiding an organization from ad hoc activity to recorded, disciplined, and consistent ways of working.

Version 3.0 is built on the CMMI Performance Solutions ecosystem (formerly the CMMI V2.0 Product Suite). Its architecture is deliberately flexible and modular so that new content can be added at the speed of business and technology change, rather than requiring a wholesale model rewrite.

Performance areas CMMI addresses

CMMI guides improvement toward business objectives in the following areas:

Cost management	Process	Schedule
Customer satisfaction	Productivity	Security
Functionality	Quality	Staff development
Organization finance	Safety	Supplier

Why use the CMMI?

CMMI helps an organization understand its current level of capability and performance. Where business needs and objectives are not being met, CMMI practices guide improvement. Three conditions are required for this to work:

- An effective and **sustainable performance improvement program** must be in place.
- Improvement must focus on **process performance and the resulting outcomes and work products**, not on paperwork.
- **Executive management must visibly and actively support** the improvement effort.

Purpose, audience, and what CMMI is not

Audience. The model is used by anyone with an interest in improving performance and processes — practitioners, managers, executive sponsors, process groups, appraisal teams, and consultants across development, services, supplier management, data, people, safety, security, and virtual work.

What CMMI is not. The model does *not* imply any specific methodology, work product type, or implementation. It is not a recipe, not a one-size-fits-all checklist, and not a set of implementable processes. Each organization must work out how to implement processes that meet the intent of the practices in its own context. Practice statements are written to be clear, unambiguous, and applicable to any context; the informative material explains how to interpret them and must not be ignored.

Two kinds of informative material

Type	What it contains
Explanatory information	Applies to **all** contexts. Helps users understand intent and business value. May include: additional explanatory information; example activities; example work products; related Practice Areas; external links (training, templates, example assets). Examples are neither prescriptive nor exhaustive.

Type	What it contains
Context specific information	Relevant to a particular industry, methodology, or discipline. Each instance has a **context tag**, a **context** statement, and explanatory information. Context tags in V3.0 include Agile Development, Data, Development, DevSecOps, People, Safety, Security, Services, and Suppliers.

The architecture also supports adding implementation guidance for domains such as information technology, cybersecurity, healthcare, telecommunications, aerospace, finance, transportation, education, government, hospitality, and consulting; and it is designed to work alongside other standards and methodologies including AS9100, Automotive SPICE, COBIT, ISO, ITIL, Kanban, Lean, and NIST.

2. Model architecture and content organization

The CMMI Model V3.0 document is organised into parts. The overview part explains the model; the Practice Areas part carries the majority of the content; the appendices carry the reference material (core PAs and Capability Areas, predefined views, adoption resources, misperceptions, glossary, abbreviations, history, references).

The component hierarchy

Component	Definition
View	A window into the model that lets an organization focus on what matters to it. Views may be predefined (domain, Capability Area, maturity level, capability level) or custom-built.
Domain	A functionally similar grouping of PAs aligned to an organization's primary capability (e.g. Development, Services). A domain is a type of view.
Category	A logical group of related Capability Areas addressing common business problems. There are four: Doing, Managing, Enabling, Improving.
Capability Area (CA)	A group of related PAs that together improve performance in a set of skills and activities. A CA is a type of view. There are eleven.
Practice Area (PA)	A set of practices that collectively describe the critical activities needed to achieve a defined intent and value. There are 31 (16 core + 15 domain-specific).
Practice Group	The organizing structure for practices within a PA. Practice group levels 1–5 form an evolutionary path of increasing capability.
Practice	The individual statement of activity, with its own required and explanatory information.

Language conventions

In the CMMI, **“or” is inclusive** — it can mean both “and” and “or”. “Manage plans and activities” means managing both. “Manage risks or opportunities” means risks, or opportunities, or both. Where a word has a special meaning it appears in the glossary; otherwise the common English dictionary meaning applies.

Each PA also carries an **identification icon** used to show relationships, support training, and act as an online selection button or tile.

3. Views, Domains, Capability Areas, and Categories

The eight Domains

A domain is an organizing principle used by both the model and the appraisal method. An organization selects one or more domains (for example, CMMI-DEV plus CMMI-SVC, adding CMMI-SEC if security matters) to form its model view.

Domain	Capability description
Data (DATA)	Governing and managing data and data quality.
Development (DEV)	Creating products or solutions, including hardware and software, and their related components.
People (PPL)	Developing, retaining, and enabling the workforce to accomplish objectives.
Safety (SAF)	Providing and maintaining safe products, services, and other solutions.
Security (SEC)	Identifying and strengthening critical defenses and increasing resilience against threats.
Services (SVC)	Building and delivering an intangible solution comprised of activities or work.
Suppliers (SPM)	Managing a company, organization, or person that supplies products, services, or other solutions.
Virtual (VRT)	Delivering products, services, or other solutions from remote locations.

The 16 Core Practice Areas

Core PAs apply regardless of domain and are included in every maturity level view.

CAR	**GOV**	**PAD**	**PR**
CM	**II**	**PCM**	**RDM**
DAR	**MC**	**PLAN**	**RSK**
EST	**MPM**	**PQA**	**VV**

The 15 Domain-Specific Practice Areas

Domain	Practice Areas
Data (DATA)	DM, DQ
Development (DEV)	PI, TS
People (PPL)	OT, WE
Safety (SAF)	ESAF
Security (SEC)	ESEC, MST
Services (SVC)	CONT, IRP, SDM, STSM
Suppliers (SPM)	SAM
Virtual (VRT)	EVW

16 core + 15 domain-specific = **31 Practice Areas** in CMMI V3.0.

The four Categories and eleven Capability Areas

Categories group related Capability Areas around the common problems businesses face when producing or delivering solutions. They exist because small, similar groupings are easier to learn, remember, and prioritise. They help an organization decide where to focus: customer satisfaction problems point to **Doing**; planning problems point to **Managing**; complexity and change point to **Enabling**; and lost improvement momentum points to **Improving**.

Category: Doing

Producing, buying, and delivering quality solutions.

Capability Area	Focus	PAs
Ensuring Quality	Covers both quality assurance and quality control.	PR, PQA, RDM, VV
Engineering and Developing Products	Engineering, developing, and delivering products and product components.	PI, TS
Delivering and Managing Services	Building the capability to deliver agreed services, deploying new/changed services, and holding a service portfolio.	SDM, STSM
Selecting and Managing Suppliers	Establishes the buyer/supplier partnership so quality solutions reach the customer and end user.	SAM

Category: Managing

Planning and managing the work and the workforce.

Capability Area	Focus	PAs
Planning and Managing Work	Sizing the work, planning and scheduling it, confirming resources are adequate, and keeping execution aligned to plan.	EST, MC, PLAN
Managing Business Resilience	Anticipating, preparing for, and responding to interruptions; identifying, evaluating, prioritizing and handling risk; keeping critical functions running.	CONT, IRP, RSK
Managing the Workforce	Developing, retaining, aligning, and empowering the people needed for current and future work.	EVW, OT, WE

Category: Enabling

Analyzing causes, making decisions, protecting the integrity of work products and data, and communicating with stakeholders.

Capability Area	Focus	PAs
Managing Data	Organizing and using data within business constraints and verifying its integrity for sound decisions.	DM, DQ
Supporting Implementation	Addressing causes of outcomes, structuring decision-making, maintaining work product integrity, and coordinating stakeholders.	CAR, CM, DAR
Managing Security and Safety	Defining security and safety strategies and functions to protect the whole ecosystem: prepare, investigate, monitor, protect and defend, preempt and prevent, review and evaluate.	ESAF, ESEC, MST

Category: Improving

Developing, managing, and improving processes and assets, focused on organizational performance.

Capability Area	Focus	PAs
Improving Performance	Measuring and understanding capability and performance, then identifying the improvement actions and assets needed.	MPM, PAD, PCM

Capability Area	Focus	PAs
Sustaining Habit and Persistence	Verifying processes are habitually and persistently performed and sustained across the organization.	GOV, II

4. Practice Areas, Practice Groups, and Practices

What a Practice Area contains

Section	Contents
Name & icon	Identifier used for relationships, training, and selection tiles.
Required PA information	**Intent** – the results and accomplishments expected from the PA. **Value** – the business value achievable by adopting its practices. **Additional required information** – further description needed to understand the intent (not present for every PA; may be view-specific).
Explanatory PA information	Practice summary; additional information; related Practice Areas (common relationships, not an exhaustive list); context specific information where applicable.
Practice groups	The organizing structure for the practices. Practice group levels show increasing capability; other groupings (by function, theme, or thread — as ISO does with clauses) are also possible within the architecture.

What a practice contains

Required practice information	Explanatory practice information
• Practice statement • Value statement (business value of the practice) • Additional required material describing scope and intent to support consistent interpretation	• Additional explanatory information • Example activities • Example work products • Related Practice Areas (as needed) • Context specific information (may be multiple instances: context identifier/description plus informative material)

Order does not imply sequence. The order of practices within a PA and within a practice group does not imply or require a sequential process. Processes meeting the intent may be performed iteratively, in parallel, or in any order that suits the business.

5. Practice Group Levels

Within each PA, practices are organised into practice group levels. Each level **builds on the levels below**, representing an increase in functionality and capability, and may add new functionality.

Level	Characteristics
Level 0 - Incomplete	An incomplete approach; the intent of the Practice Area may or may not be addressed. (Used in appraisal rating, no practices are written at this level.)
Level 1 - Initial	Processes are performed but may not be recorded. Basic practices that form an initial approach to the PA intent; not a complete set. What you expect from an organization just starting out. Begins to focus on performance issues.
Level 2 - Managed	Processes are performed against a recorded project/work-level process description. A simple but complete set of practices that fully address the PA intent. Organizational assets or standards are not required; project performance objectives are identified and monitored.
Level 3 - Defined	Processes follow a recorded organizational-level process description. Organizational standards are used and tailored to project characteristics; organizational assets are used and contributed to. Both project and organizational performance are managed.
Level 4 - Quantitatively Managed	Processes are managed and analyzed using statistical and other quantitative techniques against a recorded organizational process. Quantitative prediction of whether quality and process performance objectives will be met; special cause variation is understood.
Level 5 - Optimizing	Processes are statistically and quantitatively optimized. Statistical and quantitative techniques are used to optimize performance against business, measurement, and quality/process performance objectives. Common cause variation is understood and improved.

The essential progression, in one line each

- **Level 1** – you *do* the thing, but not from a written process, and not completely.
- **Level 2** – you do it *completely*, from a written *project*-level process.
- **Level 3** – you do it from a written *organizational* process, tailored, feeding assets back.
- **Level 4** – you do it and *predict* it statistically; special-cause variation is understood.
- **Level 5** – you *optimize* it statistically; common-cause variation is understood and attacked.

6. Maturity Levels, Capability Levels, and Views

Maturity levels

Maturity levels apply to an organization's achievement across a **predefined set of PAs** — the core PAs plus one or more selected domains. Maturity levels build on one another and **cannot be skipped**. Appraisals verify the achievement of practice group levels, which in turn determine maturity and capability level ratings.

Maturity level	Characteristics
ML 0 - Incomplete	Ad hoc and unknown. Work may or may not get completed.
ML 1 - Initial	Unpredictable and reactive. Work gets completed but is often delayed and over budget.
ML 2 - Managed	Managed at the project level. Projects are planned, performed, measured, and controlled.
ML 3 - Defined	Proactive rather than reactive. Organization-wide standards guide projects, programs, and portfolios.
ML 4 - Quantitatively Managed	Measured and controlled. The organization is data-driven, with quantitative performance improvement objectives that are predictable and aligned to stakeholder needs.
ML 5 - Optimizing	Stable and flexible. Focused on continuous improvement and built to pivot and respond to change; stability provides a platform for agility and innovation.

What each maturity level requires

To achieve	Requirement (in all targeted PAs)
Maturity Level 1	Achieve Practice Group Level 1 in all targeted PAs.
Maturity Level 2	Achieve Practice Group Level 2 in all targeted PAs (Level 1 is subsumed in Level 2).
Maturity Level 3	Achieve Practice Group Levels 2 and 3 in all targeted PAs.
Maturity Level 4	Achieve Practice Group Levels 2, 3, and 4 in all targeted PAs.
Maturity Level 5	Achieve Practice Group Levels 2, 3, 4, and 5 in all targeted PAs.

Capability levels

Capability levels apply to achievement in **individual PAs** rather than across a predefined set. Key rules:

- The **maximum capability level a PA can achieve is Capability Level 3**.
- **All capability level ratings must include II and GOV** in scope.
- A rating can be assigned to a single PA, provided II and GOV are included. For example, an organization achieves Capability Level 3 for PLAN if the practice group levels up to Level 3 are achieved in PLAN, II, and GOV.

Details of how ratings are determined live in the Method Definition Document (MDD), not the model.

Custom views

If no predefined view fits, an organization can build its own by selecting the components to include. The model gives two illustrations:

Example	PAs selected
A very small development organization just starting its improvement journey	EST, PLAN, MC, TS, PI
A technical division managing development, services, and suppliers	IRP, CONT, STSM, TS, PI, MPM, SDM, SAM, RDM, GOV, II

The method for creating a custom view: **identify the problems** → **determine root causes** → **identify potential process focal points**. And if a rating is a desired outcome, **GOV and II must be in the appraisal scope**.

7. Part Two — Successfully Adopting CMMI

CMMI is best used to address the challenges an organization is actually encountering and to continuously improve performance in the areas that matter most to it and its customers. Process improvement instils discipline in the organizational culture — in how work is perceived, done, and improved.

The four stages of process discipline

Stage	What it looks like
Stage 1	Process execution is ad hoc and undisciplined. Individuals follow their own unrecorded processes; outcomes vary and there is no systemic organizational learning. Improvement, when it happens, is unintentional.
Stage 2	There is conscious realization that processes are ad hoc and unrecorded. The need for performance improvement is recognized and basic improvement mechanisms are initiated.
Stage 3	Processes are recorded and mechanisms established to ensure fidelity of execution. Organizational support structures, including consistent senior management oversight, encourage continued use and improvement.
Stage 4	Processes and performance are clearly understood and are habitually and persistently followed.

As an organization progresses through the stages, process capability and maturity increase, and performance improves.

Thirteen keys to performance and process improvement

1. Demonstrate visible and active senior management support

Sponsorship must be continuous and consistent. Funding alone is not enough - people notice where management spends its time. If management is not seen to pay attention, no one else will.

2. Involve the people who do the work

Those who perform a task must help describe and record the process so it reflects reality. Involvement builds ownership; exclusion breeds resistance.

3. Record the 'as is' first

Record what is actually done, not what you think should be done, and resist improving the process while recording it. Capture improvement ideas separately for later analysis and prioritization.

4. Focus on meeting business objectives

Every improvement should trace to business objectives, and improvement priorities should mirror organizational priorities. Otherwise support fades.

5. Communicate, communicate, communicate

Explain what is changing, why, what help is available, how people can contribute, how success is measured, and how to give feedback. A message must be repeated in multiple ways to be received and acted on.

6. Establish a performance improvement infrastructure

Ensure funding, resources, tools, training, and support - with skilled people who have clear responsibility, authority, and accountability.

7. Target the right level of detail

Aim for process descriptions that are understandable, neither too prescriptive nor too vague. Target an '80% solution' and let continuous improvement handle the rest. Do not let perfect be the enemy of good enough.

8. Plan and provide training

Cover the full scope of processes and tools for the people doing the work; just-in-time training is a best practice. Options include classroom, mentoring, formal OJT, walkthroughs, brown bags, virtual/hybrid/e-learning.

9. Measure to achieve business results

Every measure should have a business reason. Prioritize analysis over volume of measures. Never use process measures to reward or punish individuals - doing so corrupts data quality.

10. Reinforce good behavior

Behavior change needs consistent reinforcement. Positive reinforcement generally beats punishment; base rewards on merit and performance. Address resistance with rewards, coaching/mentoring, and training.

11. Manage stakeholder expectations

Improvements ripple beyond the immediate problem; manage effects on internal and external customers and suppliers.

12. Plan for differences

What works in one context may be counterproductive in another. Processes must fit the organization's culture, including geographic and cultural differences. The litmus test is whether the improvement delivers value.

13. Recognize change can overwhelm

People absorb only so much change at once; too many simultaneous changes obscure which one worked. Allow time for habit to form - process discipline is fragile under stress.

8. Part Three — Process Habit and Persistence

Habit	A tendency or practice, especially one that is hard to give up.
Persistence	Firm or stubborn continuance in a course of action despite difficulty or opposition.

If a process is ignored, abandoned under pressure, or allowed to erode over time, it is **neither habitual nor persistent**. The goal of an improvement initiative is for the improvements to become “the way we do business” — new organizational habits.

CMMI is **not** intended to be adopted merely to pass an appraisal. Adoption is an investment, and the return should be clear performance objectives and improved business outcomes that persist.

The critical distinction: habit and persistence apply to the **organization's own business processes** — their endurance and continual improvement — **not** to CMMI practices. The Sustaining Habit and Persistence (SHP) practices apply to the processes the organization develops and uses, never to the model itself.

Governance (GOV)

GOV specifies practices for senior management in support of ways of working that are relevant and important to the organization. Visible and active management involvement is critical. Management fulfils its role by:

- Setting the strategy, direction, and expectations for performance improvement.
- Providing adequate resources for process and performance improvement.
- Ensuring processes are aligned with business needs and objectives.
- Monitoring the performance and achievements of the processes.
- Reinforcing and rewarding the development and use of processes so they continue to be used and improved.

Implementation Infrastructure (II)

II describes the infrastructure needed to build, follow, sustain, and improve processes over time. “Infrastructure” here means everything required to implement, perform, and sustain the organization's processes:

- Process descriptions.
- Resource availability aligned to need — people, tools, consumables, facilities, and time to perform.
- Funding to perform the processes.
- Training relevant to assigned responsibilities.
- Objective process evaluations confirming that work is performed as intended.

Without infrastructure, processes are not followed, sustained, or improved. Process descriptions should be clear and concise and should not demand heavy administration. A typical process description states:

Purpose	the value it delivers
Entrance criteria	when to start
Activities	what to do
Inputs and outputs	what the activities use and produce
Exit criteria	when the process is done and the value achieved

9. Part Four — Achieving High Maturity

“High maturity” means applying **statistical and other quantitative techniques** to selected processes. It is a fundamental shift in how processes are understood, managed, and improved. High maturity organizations consistently and predictively lower delivery risk while raising solution quality; they anticipate change and can pivot quickly.

What high maturity organizations do

- Establish quantitative quality and process performance objectives (QPPOs) derived from business objectives.
- Have a clear quantitative understanding of improvement ROI.
- Make data-driven decisions.
- Systematically analyze variation and understand its impact on quality and performance, giving quantitative insight into risk.
- Achieve key performance objectives more effectively.
- Understand process stability and capability and use them to manage projects and improve performance.
- Predict and achieve consistent performance.
- Increase schedule and cost performance and reduce rework.
- Focus on innovation and competitiveness, with greater confidence in measurement indicators.

The QPPO funnel

Every organization has business objectives. **Quality and Process Performance Objectives (QPPOs)** are traceable to those business and performance objectives, and they are the start of the funnel that narrows down which processes are worth the investment of quantitative management. Practice Group Level 2 and 3 practices are the **foundational building blocks** — high maturity cannot be bolted on without them.

Maturity Levels 2–3 versus 4–5

ML 2 & 3	ML 4 & 5
Processes are performed and defined; performance is managed against plans and objectives qualitatively.	Processes are managed and optimized using statistical and quantitative techniques; performance is predicted, not just tracked.
Deviation is detected and corrected after the fact.	Variation is analysed — special cause at ML 4, common cause at ML 5 — and achievement of QPPOs is forecast.

Data quality is the enabler

Poor data quality undermines the whole edifice. Its consequences include:

- Inability to conduct hypothesis tests and predictive modeling
- Inability to manage quality and performance
- Inability to meet budget and schedule
- Ineffective process changes
- Improper architecture and design decisions
- Bad information leading to bad decisions

High maturity organizations therefore:

- Collect, validate, and use high-quality data with integrity across the organization.
- Use statistical and quantitative methods to plan and manage progress against objectives.

- Construct Process Performance Baselines (PPBs) to manage work and target improvement areas.
- Develop Process Performance Models (PPMs) to understand relationships among processes and sub-processes.
- Evaluate the impact of proposed improvements on achieving QPPOs.
- Improve quality, schedule, and cost simultaneously with a quantitative understanding of the trade-offs.

A worked illustration: should we accept the work?

A customer asks for a feature in 10 weeks. Historically, similar features have taken 9 to 11 weeks. The decision does not rest on the range — it rests on the **distribution**. If most of the historical distribution sits at ten weeks or more, the commitment is unsafe; if most of it sits below ten weeks, the commitment is sound. That is the difference high maturity makes.

Understanding variation

A statistical measure of variation includes a summary or characterization of the distribution, a characterization of **central tendency** (mean, median, mode), and a characterization of **dispersion** (variance, standard deviation, range).

Improvements can **shift the mean**, **reduce the variation**, or both — two different insights into process performance. High maturity organizations understand the balance between stability and change: they can predict the performance impact and ROI of a change before making it, and measure its effect afterwards.

10. Practice Area Reference

All 31 Practice Areas, alphabetically by abbreviation grouping used in the source. For each PA: its Category and Capability Area, its **Intent** and **Value** (the required information), a note, and every practice organised by practice group level.

PA	Name	Category / Capability Area	Max level
CAR	Causal Analysis and Resolution	Enabling / Supporting Implementation	5
CM	Configuration Management	Enabling / Supporting Implementation	2
CONT	Continuity	Managing / Managing Business Resilience	3
DM	Data Management	Enabling / Managing Data	3
DQ	Data Quality	Enabling / Managing Data	3
DAR	Decision Analysis and Resolution	Enabling / Supporting Implementation	3
ESAF	Enabling Safety	Enabling / Managing Security and Safety	3
ESEC	Enabling Security	Enabling / Managing Security and Safety	3
EVW	Enabling Virtual Work	Managing / Managing the Workforce	3
EST	Estimating	Managing / Planning and Managing Work	3
GOV	Governance	Improving / Sustaining Habit and Persistence	4
II	Implementation Infrastructure	Improving / Sustaining Habit and Persistence	4
IRP	Incident Resolution and Prevention	Managing / Managing Business Resilience	3
MPM	Managing Performance and Measurement	Improving / Improving Performance	5
MST	Managing Security Threats and Vulnerabilities	Enabling / Managing Security and Safety	4
MC	Monitor and Control	Managing / Planning and Managing Work	3
OT	Organizational Training	Managing / Managing the Workforce	3
PR	Peer Reviews	Doing / Ensuring Quality	3
PLAN	Planning	Managing / Planning and Managing Work	4
PAD	Process Asset Development	Improving / Improving Performance	3
PCM	Process Management	Improving / Improving Performance	4
PQA	Process Quality Assurance	Doing / Ensuring Quality	3
PI	Product Integration	Doing / Engineering and Developing Products	3
RDM	Requirements Development and Management	Doing / Ensuring Quality	3
RSK	Risk and Opportunity Management	Managing / Managing Business Resilience	3
SDM	Service Delivery Management	Doing / Delivering and Managing Services	3
STSM	Strategic Service Management	Doing / Delivering and Managing Services	3
SAM	Supplier Agreement Management	Doing / Selecting and Managing Suppliers	4
TS	Technical Solution	Doing / Engineering and Developing Products	3

PA	Name	Category / Capability Area	Max level
VV	Verification and Validation	Doing / Ensuring Quality	3
WE	Workforce Empowerment	Managing / Managing the Workforce	3

CAR — Causal Analysis and Resolution

ENABLING / SUPPORTING IMPLEMENTATION

Intent	Identify the causes of selected outcomes and act either to prevent undesirable outcomes recurring or to make positive outcomes recur.
Value	Attacks the causes of issues, eliminating rework and directly improving quality and productivity.

Level	ID	Practice
1	**CAR 1.1**	Identify and address causes of selected outcomes.
2	**CAR 2.1**	Select the outcomes to be analyzed.
	CAR 2.2	Analyze the causes of those outcomes and address them.
3	**CAR 3.1**	Determine causes by following an organizational process.
	CAR 3.2	Propose actions that address the identified causes.
	CAR 3.3	Implement the selected action proposals.
	CAR 3.4	Record causal analysis and resolution data.
	CAR 3.5	Submit improvement proposals for changes proven effective.
4	**CAR 4.1**	Perform root cause analysis using statistical and other quantitative techniques.
	CAR 4.2	Quantitatively evaluate the effect of implemented actions on process performance.
5	**CAR 5.1**	Use quantitative analysis to decide whether a resolution should be applied to other solutions and processes to optimize organizational performance.

Note. Activities include: determining causes of selected outcomes; proposing actions; implementing selected proposals; recording the data; and submitting improvement proposals for proven changes. Preventing defects is more cost-effective than detecting them later. Because causal analysis on every outcome is impractical, targets are selected by weighing estimated investment against return.

CM — Configuration Management

ENABLING / SUPPORTING IMPLEMENTATION

Intent	Manage the integrity of work products through configuration identification, version control, change control, and audits.
Value	Reduces loss of work and increases the ability to deliver the correct version of the solution to the customer.

Level	ID	Practice
1	**CM 1.1**	Perform version control.
2	**CM 2.1**	Identify the items to be placed under configuration management.
	CM 2.2	Develop, keep updated, and use a configuration and change management system.
	CM 2.3	Develop or release baselines for internal use or customer delivery.
	CM 2.4	Manage changes to items under configuration management.
	CM 2.5	Develop, keep updated, and use records describing the items under configuration management.
	CM 2.6	Perform configuration audits to maintain the integrity of baselines, changes, and the CM system content.

Note. Configuration management protects the integrity of baselines and the ability to reproduce and release the correct version.

CONT — Continuity

MANAGING / MANAGING BUSINESS RESILIENCE

Intent	Anticipate and address disruptions to critical business operations so work can continue or resume as quickly as possible.
Value	Enables continued operation when serious disruptions or catastrophic events occur.

Level	ID	Practice
1	**CONT 1.1**	Develop contingency approaches for managing significant disruptions.
2	**CONT 2.1**	Identify and prioritize functions essential for continuity.
	CONT 2.2	Identify and prioritize resources essential for continuity.
	CONT 2.3	Develop, keep updated, and follow continuity plans to resume essential functions.
3	**CONT 3.1**	Develop and keep updated continuity training materials.
	CONT 3.2	Provide and evaluate continuity training according to plan.
	CONT 3.3	Prepare, conduct, and analyze verification and validation of the continuity plan.

Note. Continuity plans and validates the minimum critical set of functions and resources needed to keep operating through significant or catastrophic events.

DM — Data Management

ENABLING / MANAGING DATA

Intent	Identify, implement, and control the approach and activities used to manage data.
Value	Maximizes operational efficiency by prioritizing the critical data activities needed to meet performance objectives.

Level	ID	Practice
1	**DM 1.1**	Identify data management objectives.
	DM 1.2	Use metadata to manage data.
2	**DM 2.1**	Develop, keep updated, and follow a data management approach aligned to objectives.
	DM 2.2	Establish a data management architecture that supports the approach.
3	**DM 3.1**	Establish and deploy an organizational data management capability.
	DM 3.2	Periodically review the effectiveness of the organization's data management capability and act on the results.

Note. Covers the processes, techniques, and tools for collecting, keeping, and using data securely, efficiently, and cost-effectively.

DQ — Data Quality

ENABLING / MANAGING DATA

Intent	Develop, follow, and keep updated an approach for implementing data quality standards.
Value	Maximizes the value and accuracy of data for effective operations and consistent decision-making.

Level	ID	Practice
1	**DQ 1.1**	Identify data quality parameters.
	DQ 1.2	Perform data cleansing activities.
2	**DQ 2.1**	Define criteria for data cleansing.
	DQ 2.2	Develop, keep updated, and follow a data quality approach.
	DQ 2.3	Perform data cleansing based on the criteria and the data quality approach.
3	**DQ 3.1**	Conduct data quality assessments.
	DQ 3.2	Periodically review the effectiveness of the organization's data quality activities and act on the results.

Note. Data quality standards may come from external regulations or internal requirements; the approach must cover all sources and the full data lifecycle, and address consistency.

DAR — Decision Analysis and Resolution

ENABLING / SUPPORTING IMPLEMENTATION

Intent	Make and record decisions using a recorded process that analyzes alternatives.
Value	Increases the objectivity of decision-making and the probability of selecting the optimal solution.

Level	ID	Practice
1	**DAR 1.1**	Define and record the alternatives.
	DAR 1.2	Make and record the decision.
2	**DAR 2.1**	Develop, keep updated, and use rules for when a recorded, criteria-based decision process must be followed.
	DAR 2.2	Develop criteria for evaluating alternatives.
	DAR 2.3	Identify alternative solutions.
	DAR 2.4	Select evaluation methods.
	DAR 2.5	Evaluate and select solutions using the criteria and methods.
3	**DAR 3.1**	Develop, keep updated, and use a description of role-based decision authority.

Note. Not every decision warrants a formal analysis - rules determine which decisions require the recorded, criteria-based process.

ESAF — Enabling Safety

ENABLING / MANAGING SECURITY AND SAFETY

Intent	Minimize and mitigate safety risks within the tolerance parameters and constraints of operational effectiveness, time, and cost.
Value	Reduces residual safety hazard risk to an acceptable tolerance level.

Level	ID	Practice
1	**ESAF 1.1**	Identify and record safety needs and hazards.
	ESAF 1.2	Address prioritized safety needs and hazards.

Level	ID	Practice
2	**ESAF 2.1**	Identify critical safety needs and constraints, keep them updated, and use them to set and maintain safety objectives.
	ESAF 2.2	Develop, keep updated, and follow an approach for workplace environment safety.
	ESAF 2.3	Develop, keep updated, and follow an approach for functional safety of the solution.
3	**ESAF 3.1**	Establish and deploy an organizational safety capability.
	ESAF 3.2	Perform safety evaluations periodically and act on the results.
	ESAF 3.3	Develop, keep updated, and follow organizational safety control strategies.

Note. Covers safety across solutions, products, processes, services and environments - both facilitating and managing safety activities.

ESEC — Enabling Security

ENABLING / MANAGING SECURITY AND SAFETY

Intent	Develop and keep updated a security approach that anticipates, identifies, and acts to avoid or minimize the impact of security issues on the organization or solution.
Value	Reduces the impact of security threats and vulnerabilities on business performance.

Level	ID	Practice
1	**ESEC 1.1**	Identify and record security needs and issues.
	ESEC 1.2	Address prioritized security needs and issues.
2	**ESEC 2.1**	Identify and record security needs, keep them updated, and use them to develop a security approach and objectives.
	ESEC 2.2	Develop, keep updated, and follow an approach for physical security needs.
	ESEC 2.3	Develop, keep updated, and follow an approach for mission, personnel, and process-related security needs.
	ESEC 2.4	Develop, keep updated, and follow an approach for cybersecurity needs.
3	**ESEC 3.1**	Establish and deploy an organizational security operations capability.
	ESEC 3.2	Develop, follow, and keep updated an organizational security strategy, approach, and architecture.
	ESEC 3.3	Periodically perform organization-wide security reviews and evaluations and act on the results.

Note. Security is a continuous (24/7/365) activity, not an afterthought and not a trade-off item against schedule, cost, or quality.

EVW — Enabling Virtual Work

MANAGING / MANAGING THE WORKFORCE

Intent	Define and manage an approach for effective virtual work and operations.
Value	Maximizes delivery effectiveness and efficiency while reducing the impact and expense of travel and in-person activities.

Level	ID	Practice
1	**EVW 1.1**	Identify and record virtual work needs and constraints.
	EVW 1.2	Perform virtual work.
2	**EVW 2.1**	Develop, keep updated, and use an approach for performing virtual work.
	EVW 2.2	Monitor the virtual work approach and take corrective action when needed.
3	**EVW 3.1**	Develop, keep updated, and use an organizational strategy, approach, and functional capability for virtual work.
	EVW 3.2	Periodically review the effectiveness of the organization's virtual work approach and act on the results.

Note. Addresses virtual, remote, and hybrid work needs including personnel, process, technical, and security considerations.

EST — Estimating

MANAGING / PLANNING AND MANAGING WORK

Intent	Estimate the size, effort, duration, and cost of the work and the resources needed to develop, acquire, or deliver the solution.
Value	Provides a basis for commitments and planning, reduces uncertainty, enables early corrective action, and increases the likelihood of meeting objectives.

Level	ID	Practice
1	**EST 1.1**	Develop high-level estimates for performing the work.
2	**EST 2.1**	Develop, keep updated, and use the scope of what is being estimated.
	EST 2.2	Develop and keep updated size estimates for the solution.
	EST 2.3	Based on size, develop and record effort, duration, and cost estimates with their rationale.
3	**EST 3.1**	Develop and keep updated a recorded estimation method.
	EST 3.2	Use the organizational measurement repository and process assets when estimating.

Note. Size is the foundation: effort, duration, and cost estimates are derived from it, and rationale is recorded.

GOV — Governance

IMPROVING / SUSTAINING HABIT AND PERSISTENCE

Intent	Guide senior management in their role sponsoring and governing performance, processes, and related activities.
Value	Minimizes the cost of process implementation, increases the likelihood of meeting objectives, and verifies that processes support business success.

Level	ID	Practice
1	**GOV 1.1**	Senior management identifies what is important for doing the work and defines the approach needed to meet organizational objectives.

Level	ID	Practice
2	GOV 2.1	Senior management defines, keeps updated, and communicates organizational directives for process implementation and performance improvement.
	GOV 2.2	Senior management provides funding, resources, and training to develop, support, perform, improve, and evaluate adherence to processes.
	GOV 2.3	Senior management identifies its information needs and uses the collected information for governance and oversight.
	GOV 2.4	Senior management assigns authority and holds people accountable for adhering to directives and achieving process and performance objectives.
3	GOV 3.1	Senior management ensures measures supporting objectives are collected, analyzed, and used throughout the organization.
	GOV 3.2	Senior management ensures competencies and processes align with organizational objectives.
4	GOV 4.1	Senior management verifies that selected decisions are driven by statistical and quantitative analysis of performance and achievement of quality and process performance objectives.

Note. GOV practices apply to the organization's own processes and their improvement - not to the CMMI practices themselves. Management sets strategy and expectations, provides resources, aligns processes with business needs, monitors achievement, and reinforces and rewards process use.

II — Implementation Infrastructure

IMPROVING / SUSTAINING HABIT AND PERSISTENCE

Intent	Ensure that the processes and assets important to organizational performance are habitually and persistently followed, used, and improved.
Value	Sustains the ability to consistently achieve goals and objectives efficiently and effectively.

Level	ID	Practice
1	II 1.1	Perform processes that address the intent of the Level 1 practices.
2	II 2.1	Provide sufficient resources, funding, and training to develop and perform processes.
	II 2.2	Develop and keep processes updated and verify they are being followed.
3	II 3.1	Use organizational processes and assets to plan, manage, and perform the work.
	II 3.2	Evaluate adherence to, and effectiveness of, the organizational processes.
	II 3.3	Contribute process-related information or assets back to the organization.
4	II 4.1	Build the organizational capability to understand and apply statistical and other quantitative techniques.

Note. 'Infrastructure' means everything needed to implement, perform, and sustain processes: process descriptions, resources, funding, training, and objective evaluations. Process descriptions typically state purpose (value), entrance criteria, activities, inputs and outputs, and exit criteria. II practices apply to the organization's processes, not to CMMI practices.

IRP — Incident Resolution and Prevention

MANAGING / MANAGING BUSINESS RESILIENCE

Intent	Resolve and prevent disruptions promptly to sustain service delivery levels.
Value	Minimizes the impact of disruptions so objectives and customer commitments are met more effectively.

Level	ID	Practice
1	**IRP 1.1**	Record and resolve incidents.
2	**IRP 2.1**	Develop, keep updated, and follow an approach for incident resolution and prevention.
	IRP 2.2	Monitor and resolve each incident to closure.
	IRP 2.3	Communicate incident status.
3	**IRP 3.1**	Develop, keep updated, and use an incident management system to process and track incidents and resolutions.
	IRP 3.2	Analyze selected incident and resolution data to prevent future incidents.

Note. Covers both actual and potential incidents that may affect operations and solution delivery, and minimizing their recurrence.

MPM — Managing Performance and Measurement

IMPROVING / IMPROVING PERFORMANCE

Intent	Manage performance using measurement and analysis in order to achieve business objectives.
Value	Maximizes business return on investment by focusing management and improvement effort on cost, schedule, and quality performance.

Level	ID	Practice
1	**MPM 1.1**	Collect measures and record performance.
	MPM 1.2	Identify and address performance issues.
2	**MPM 2.1**	Derive, record, and keep updated measurement and performance objectives from selected business needs and objectives.
	MPM 2.2	Develop, keep updated, and use operational definitions for measures.
	MPM 2.3	Obtain measurement data according to the operational definitions.
	MPM 2.4	Analyze performance and measurement data according to the operational definitions.
	MPM 2.5	Store measurement data, specifications, and analysis results according to the operational definitions.
	MPM 2.6	Act on identified issues with meeting measurement and performance objectives.
3	**MPM 3.1**	Develop, keep updated, and use organizational measurement and performance objectives traceable to business objectives.
	MPM 3.2	Follow organizational processes and standards for developing and using operational definitions.
	MPM 3.3	Develop, keep updated, and follow a data quality process.
	MPM 3.4	Develop, keep updated, and use the organization's measurement repository.
	MPM 3.5	Analyze organizational performance to determine and address improvement needs.
	MPM 3.6	Periodically communicate performance results across the organization.

Level	ID	Practice
4	**MPM 4.1**	Use quantitative techniques to develop, keep updated, and communicate quality and process performance objectives (QPPOs) traceable to business objectives.
	MPM 4.2	Select measures and analytic techniques to quantitatively manage performance toward QPPOs.
	MPM 4.3	Use quantitative techniques to develop, analyze, and keep updated process performance baselines.
	MPM 4.4	Use quantitative techniques to develop, analyze, and keep updated process performance models.
	MPM 4.5	Use quantitative techniques to determine or predict achievement of QPPOs.
5	**MPM 5.1**	Use quantitative techniques to ensure business objectives align with business strategy to optimize performance.
	MPM 5.2	Analyze performance data quantitatively to determine the organization's ability to satisfy selected business objectives and identify improvements.
	MPM 5.3	Select and implement improvement proposals based on quantitative analysis of their expected effect on meeting and optimizing objectives.

Note. The largest PA in the model and the backbone of high maturity: it links business objectives to measures, baselines, models, and improvement decisions.

MST — Managing Security Threats and Vulnerabilities

ENABLING / MANAGING SECURITY AND SAFETY

Intent	Identify security threats and vulnerabilities that could compromise the organization or solution, analyze their potential impact, and define and take mitigating action.
Value	Increases the organization's capability and resilience to identify, mitigate, and recover from threats and vulnerabilities.

Level	ID	Practice
1	**MST 1.1**	Identify and record security threats and vulnerabilities.
	MST 1.2	Take action to address security threats and vulnerabilities.
2	**MST 2.1**	Develop, keep updated, and follow an approach for handling threats and vulnerabilities.
	MST 2.2	Develop and keep updated criteria for evaluating threats and vulnerabilities.
	MST 2.3	Use the recorded criteria to prioritize, monitor, and address the most critical threats and vulnerabilities arising in operations.
	MST 2.4	Evaluate and report the effectiveness of the approach and the actions taken.
3	**MST 3.1**	Develop, keep updated, and follow an organizational security strategy, approach, and architecture to evaluate, manage, and verify threats and vulnerabilities.
	MST 3.2	Analyze security verification and validation results for accuracy, comparability, consistency, and validity across the organization.
	MST 3.3	Evaluate the effectiveness of the organizational security strategy, approach, and architecture.
4	**MST 4.1**	Employ threat intelligence analysis to improve the security approach and architecture and to select security solutions.

Note. A holistic, systematic approach to selecting which threats and vulnerabilities are most critical given risk and impact to the business, mission, or solution.

MC — Monitor and Control

MANAGING / PLANNING AND MANAGING WORK

Intent	Provide an understanding of project progress so that appropriate corrective action can be taken when performance deviates significantly from plan.
Value	Increases the probability of meeting objectives by acting early on significant deviations.

Level	ID	Practice
1	**MC 1.1**	Record task completions.
	MC 1.2	Identify and resolve issues.
2	**MC 2.1**	Track actual results against estimates for size, effort, schedule, resources, knowledge and skills, and budget.
	MC 2.2	Track the involvement of identified stakeholders and their commitments.
	MC 2.3	Monitor the transition to operations and support.
	MC 2.4	Take corrective action when actuals differ significantly from plan and manage the actions to closure.
3	**MC 3.1**	Manage the project using the project plan and the project process.
	MC 3.2	Manage critical dependencies and activities.
	MC 3.3	Monitor the work environment to identify issues.
	MC 3.4	Manage and resolve issues with affected stakeholders.

Note. The control counterpart to PLAN and EST: it compares actuals to plan and drives corrective action.

OT — Organizational Training

MANAGING / MANAGING THE WORKFORCE

Intent	Develop the skills and knowledge of personnel so they perform their roles efficiently and effectively.
Value	Enhances individual skills and knowledge to improve organizational work performance.

Level	ID	Practice
1	**OT 1.1**	Train people.
2	**OT 2.1**	Identify training needs.
	OT 2.2	Train personnel and keep records.
3	**OT 3.1**	Develop and keep updated the organization's strategic and short-term training needs.
	OT 3.2	Coordinate training needs and delivery between projects and the organization.
	OT 3.3	Develop, keep updated, and follow strategic and short-term training plans.
	OT 3.4	Develop, keep updated, and use a training capability that addresses organizational needs.
	OT 3.5	Assess and report the effectiveness of the organization's training program.
	OT 3.6	Record, keep updated, and use the set of organizational training records.

Note. Provides a strategy and capability for training that supports strategic business objectives and common tactical needs.

PR — Peer Reviews

DOING / ENSURING QUALITY

Intent	Identify and address process performance and work product issues through review by the producer's peers or subject matter experts.
Value	Reduces cost and rework by uncovering issues or defects early.

Level	ID	Practice
1	**PR 1.1**	Perform reviews of work products and record issues.
2	**PR 2.1**	Develop and keep updated the procedures and supporting materials used to prepare for and perform peer reviews.
	PR 2.2	Select the work products to be peer reviewed.
	PR 2.3	Prepare and perform peer reviews using the established procedures.
	PR 2.4	Resolve the issues identified in peer reviews.
3	**PR 3.1**	Analyze results and data from peer reviews.

Note. Peer reviews find defects close to their point of insertion, which is where they are cheapest to fix.

PLAN — Planning

MANAGING / PLANNING AND MANAGING WORK

Intent	Develop plans that describe what is needed to accomplish the work within the standards and constraints of the organization.
Value	Optimizes cost, functionality, and quality, increasing the likelihood of meeting objectives.

Level	ID	Practice
1	**PLAN 1.1**	Develop a list of tasks.
	PLAN 1.2	Assign people to tasks.
2	**PLAN 2.1**	Develop and keep updated the approach for accomplishing the work.
	PLAN 2.2	Plan for the knowledge and skills needed.
	PLAN 2.3	Using recorded estimates, develop and keep updated the budget and schedule.
	PLAN 2.4	Plan the involvement of identified stakeholders.
	PLAN 2.5	Plan the transition to operations and support.
	PLAN 2.6	Ensure plans are feasible by reconciling estimates against resource capacity and availability.
	PLAN 2.7	Develop the project plan, ensure internal consistency, and keep it updated.
	PLAN 2.8	Review plans and obtain commitment from affected stakeholders.
3	**PLAN 3.1**	Use the organization's standard processes and tailoring guidelines to develop, keep updated, and follow the project process.
	PLAN 3.2	Develop and keep updated a plan using the project process, organizational assets, and the measurement repository.
	PLAN 3.3	Identify and negotiate critical dependencies.
	PLAN 3.4	Plan and keep updated the project environment based on organizational standards.
4	**PLAN 4.1**	Use statistical and other quantitative techniques to develop and keep updated project processes that enable achievement of quality and process performance objectives.

Note. Planning converts estimates into a feasible, committed plan - budget, schedule, resources, stakeholders, and transition.

PAD — Process Asset Development

IMPROVING / IMPROVING PERFORMANCE

Intent	Develop and keep updated the process assets needed to perform the work.
Value	Provides the capability to understand and repeat successful performance.

Level	ID	Practice
1	**PAD 1.1**	Develop process assets to perform the work.
2	**PAD 2.1**	Determine which process assets are needed.
	PAD 2.2	Develop, buy, or reuse process assets.
	PAD 2.3	Make processes and assets available.
3	**PAD 3.1**	Develop, keep updated, and follow a strategy for building and updating process assets.
	PAD 3.2	Develop, record, and keep updated a process architecture describing the structure of the organization's processes and assets.
	PAD 3.3	Develop, keep updated, and make available the organization's processes and assets in a process asset library.
	PAD 3.4	Develop, keep updated, and use tailoring criteria and guidelines.
	PAD 3.5	Develop, keep updated, and make available work environment standards.
	PAD 3.6	Develop, keep updated, and make available organizational measurement and analysis standards.

Note. PAD builds the organizational standard processes, process architecture, tailoring guidelines, and the process asset library.

PCM — Process Management

IMPROVING / IMPROVING PERFORMANCE

Intent	Manage and implement continuous improvement of processes and infrastructure to meet business objectives, and make performance results visible, accessible, and sustainable.
Value	Ensures processes, infrastructure, and their improvement contribute to meeting business objectives.

Level	ID	Practice
1	**PCM 1.1**	Develop a support structure that provides process guidance, fixes process problems, and drives continuous improvement.
	PCM 1.2	Appraise current process implementation and identify strengths and weaknesses.
	PCM 1.3	Address improvement opportunities and process issues.
2	**PCM 2.1**	Identify improvements to processes and process assets.
	PCM 2.2	Develop, keep updated, and follow plans for implementing selected improvements.

Level	ID	Practice
3	**PCM 3.1**	Develop, keep updated, and use process improvement objectives traceable to business objectives.
	PCM 3.2	Identify the processes that contribute most to meeting business objectives.
	PCM 3.3	Explore and evaluate potential new processes, techniques, methods, and tools.
	PCM 3.4	Support the implementation, deployment, and sustainment of improvements.
	PCM 3.5	Deploy organizational standard processes and assets.
	PCM 3.6	Evaluate and report the effectiveness of deployed improvements against improvement objectives.
4	**PCM 4.1**	Use statistical and other quantitative techniques to validate selected improvements against expectations and business objectives.

Note. PCM is the engine of the improvement cycle: appraise, select, plan, deploy, and evaluate improvements.

PQA — Process Quality Assurance

DOING / ENSURING QUALITY

Intent	Verify and enable improvement of the quality of the processes performed and the work products produced.
Value	Increases consistent use and improvement of processes to maximize business benefit and customer satisfaction.

Level	ID	Practice
1	**PQA 1.1**	Identify and address process and work product issues.
2	**PQA 2.1**	Develop, keep updated, and follow a quality assurance approach and plan based on historical quality data.
	PQA 2.2	Objectively evaluate selected performed processes and work products against the recorded process and applicable standards.
	PQA 2.3	Communicate quality and non-compliance issues and ensure they are resolved.
	PQA 2.4	Record and use the results of quality assurance activities.
3	**PQA 3.1**	Identify and record improvement opportunities during quality assurance activities.

Note. PQA provides objective evaluation - independence of judgement matters more than organizational independence.

PI — Product Integration

DOING / ENGINEERING AND DEVELOPING PRODUCTS

Intent	Integrate and deliver a solution that addresses functionality, performance, and quality requirements.
Value	Increases customer satisfaction by delivering a solution that meets or exceeds functionality and quality requirements.

Level	ID	Practice
1	**PI 1.1**	Assemble solutions and deliver them to the customer.

Level	ID	Practice
2	**PI 2.1**	Develop, keep updated, and follow an integration strategy.
	PI 2.2	Develop, keep updated, and use the integration environment.
	PI 2.3	Develop, keep updated, and follow procedures and criteria for integrating solutions and components.
	PI 2.4	Before integration, confirm each component is properly identified and operates per its requirements and design.
	PI 2.5	Evaluate integrated components for conformance to the solution's requirements and design.
	PI 2.6	Integrate solutions and components according to the integration strategy.
3	**PI 3.1**	Review and keep updated interface or connection descriptions for coverage, completeness, and consistency over the solution's life.
	PI 3.2	Before integration, confirm component interfaces or connections comply with their descriptions.
	PI 3.3	Evaluate integrated components for interface or connection compatibility.

Note. Integration is driven by a strategy, an environment, and criteria - with interfaces and connections as the main source of risk.

RDM — Requirements Development and Management

DOING / ENSURING QUALITY

Intent	Elicit requirements, confirm a common stakeholder understanding, and keep requirements, plans, and work products aligned.
Value	Increases the likelihood that the solution meets or exceeds customer expectations and needs.

Level	ID	Practice
1	**RDM 1.1**	Record requirements.
2	**RDM 2.1**	Elicit stakeholder needs, expectations, constraints, and interfaces or connections, and confirm understanding of the requirements.
	RDM 2.2	Transform stakeholder needs into prioritized customer requirements.
	RDM 2.3	Obtain commitment from project participants that they can implement the requirements.
	RDM 2.4	Develop, record, and keep updated bidirectional traceability among requirements and activities or work products.
	RDM 2.5	Ensure plans and work products remain consistent with requirements.
3	**RDM 3.1**	Develop and keep updated requirements for the solution and its components.
	RDM 3.2	Develop operational concepts and scenarios.
	RDM 3.3	Allocate the requirements to be implemented.
	RDM 3.4	Identify, develop, and keep updated interface or connection requirements.
	RDM 3.5	Ensure requirements are necessary and sufficient.
	RDM 3.6	Balance stakeholder needs and constraints.
	RDM 3.7	Validate requirements to ensure the resulting solution will perform as intended in the target environment.

Note. RDM merges requirements development and requirements management: elicitation, transformation, commitment, traceability, and validation.

RSK — Risk and Opportunity Management

MANAGING / MANAGING BUSINESS RESILIENCE

Intent	Identify, record, analyze, and manage potential risks and opportunities.
Value	Mitigates adverse impacts or capitalizes on positive impacts, increasing the likelihood of meeting objectives.

Level	ID	Practice
1	**RSK 1.1**	Identify, record, and keep updated risks or opportunities.
2	**RSK 2.1**	Analyze identified risks or opportunities.
	RSK 2.2	Monitor risks or opportunities and communicate status to affected stakeholders.
3	**RSK 3.1**	Identify and use risk or opportunity categories.
	RSK 3.2	Define and use parameters for risk or opportunity analysis and handling.
	RSK 3.3	Develop and keep updated a risk or opportunity management strategy.
	RSK 3.4	Develop and keep updated risk or opportunity management plans.
	RSK 3.5	Manage risks or opportunities by implementing the planned activities.

Note. CMMI V3.0 treats opportunity as the mirror image of risk - the same identification, analysis, and handling machinery applies to both.

SDM — Service Delivery Management

DOING / DELIVERING AND MANAGING SERVICES

Intent	Deliver services and manage the service delivery system.
Value	Increases customer satisfaction by delivering services that meet or exceed expectations.

Level	ID	Practice
1	**SDM 1.1**	Use the service system to deliver services.
2	**SDM 2.1**	Develop, record, keep updated, and follow service agreements.
	SDM 2.2	Receive and process service requests in accordance with service agreements.
	SDM 2.3	Deliver services in accordance with service agreements.
	SDM 2.4	Analyze existing service agreements and service data to prepare for new or updated agreements.
	SDM 2.5	Develop, record, keep updated, and follow the approach for operating and changing the service system.
	SDM 2.6	Confirm the service system is ready to support service delivery.
3	**SDM 3.1**	Develop, record, keep updated, and use organizational standard service systems and agreements.

Note. Service agreements are the contract between provider and customer; the service system is what makes delivery repeatable.

STSM — Strategic Service Management

DOING / DELIVERING AND MANAGING SERVICES

Intent	Develop and deploy standard services compatible with strategic business needs and plans.
Value	Increases the likelihood of meeting business objectives by aligning standard services with customer needs.

Level	ID	Practice
1	**STSM 1.1**	Develop a list of current services.
2	**STSM 2.1**	Develop, keep updated, and use descriptions of current services.
	STSM 2.2	Collect, record, and analyze data on strategic needs and service delivery capabilities.
	STSM 2.3	Develop, keep updated, and follow an approach for providing new or changed services derived from strategic needs and capabilities.
3	**STSM 3.1**	Develop, keep updated, and use the set of organizational standard service descriptions and service levels.

Note. STSM is the portfolio-level counterpart to SDM: which services to offer, and at what levels.

SAM — Supplier Agreement Management

DOING / SELECTING AND MANAGING SUPPLIERS

Intent	Select qualified suppliers, establish agreements, and manage supplier and acquirer activities over the life of the agreement.
Value	Maximizes the probability of mutual success for acquirers and suppliers.

Level	ID	Practice
1	**SAM 1.1**	Identify, evaluate, and select suppliers.
	SAM 1.2	Develop and record the supplier agreement.
	SAM 1.3	Accept or reject supplier deliverables.
	SAM 1.4	Process supplier invoices.
2	**SAM 2.1**	Identify evaluation criteria and potential suppliers, and distribute supplier requests.
	SAM 2.2	Evaluate supplier responses against the recorded criteria and select suppliers.
	SAM 2.3	Manage supplier activities as specified in the agreement and keep the agreement updated.
	SAM 2.4	Verify the supplier agreement is satisfied before accepting a deliverable.
	SAM 2.5	Manage supplier invoices according to the agreement.
3	**SAM 3.1**	Conduct technical reviews of supplier performance activities and selected deliverables.
	SAM 3.2	Manage supplier performance and processes based on criteria in the agreement.
4	**SAM 4.1**	Select measures and apply analytical techniques to quantitatively manage suppliers against performance targets.

Note. Note that SAM has four Level 1 practices - more than any other PA - because even ad hoc acquisition involves selection, agreement, acceptance, and payment.

TS — Technical Solution

DOING / ENGINEERING AND DEVELOPING PRODUCTS

Intent	Design and build solutions that meet requirements.
Value	Provides a cost-effective design and solution that meets customer requirements and reduces rework.

Level	ID	Practice
1	**TS 1.1**	Build a solution that meets requirements.
2	**TS 2.1**	Design and build a solution that meets requirements.
	TS 2.2	Evaluate the design and address identified issues.
	TS 2.3	Provide guidance on use of the solution.
3	**TS 3.1**	Develop criteria for design decisions.
	TS 3.2	Develop alternative solutions for selected components.
	TS 3.3	Perform build, buy, or reuse analysis.
	TS 3.4	Select solutions based on the design criteria.
	TS 3.5	Develop, keep updated, and use the information needed to implement the design.
	TS 3.6	Design solution interfaces or connections using established criteria.

Note. TS pairs with DAR at Level 3: design decisions become criteria-based selections among alternatives.

VV — Verification and Validation

DOING / ENSURING QUALITY

Intent	Confirm that selected solutions and components meet their requirements (verification), and demonstrate that they fulfil their intended use in the target environment (validation).
Value	Increases the likelihood that the solution will satisfy the customer.

Level	ID	Practice
1	**VV 1.1**	Perform verification to confirm requirements are implemented; record and communicate results.
	VV 1.2	Perform validation to confirm the solution will function as intended in its target environment; record and communicate results.
2	**VV 2.1**	Select components and methods for verification and validation.
	VV 2.2	Develop, keep updated, and use the environment needed to support verification and validation.
	VV 2.3	Develop, keep updated, and follow procedures for verification and validation.
3	**VV 3.1**	Develop, keep updated, and use criteria for verification and validation.
	VV 3.2	Analyze and communicate verification and validation results.

Note. Verification asks 'did we build it right?'; validation asks 'did we build the right thing?'

WE — Workforce Empowerment

MANAGING / MANAGING THE WORKFORCE

Intent	Align the workforce to business objectives and empower individuals and workgroups to perform their roles efficiently and effectively.
Value	Enhances the capability of the workforce to contribute to business success.

Level	ID	Practice
1	**WE 1.1**	Identify and allocate commitments to workgroups.
2	**WE 2.1**	Record, allocate, and keep updated work assignments based on assessed qualifications, skills, and related criteria.
	WE 2.2	Manage the transition of individuals into and out of roles and workgroups.
	WE 2.3	Develop, keep updated, and use communication and coordination mechanisms within and across workgroups.
3	**WE 3.1**	Develop, keep updated, and use workforce competencies to build organizational capability and achieve objectives.
	WE 3.2	Develop, keep updated, and use an organizational structure and approach that empowers workgroups.
	WE 3.3	Develop, keep updated, and use organizational compensation strategies and mechanisms.

Note. WE covers assignment, transition, coordination, competencies, empowering structures, and compensation.

11. Glossary of Key Terms

Terms take on the character of the content where they appear: a term used in required information is required in that context; a term in explanatory information is explanatory in that context. Where a word has no special CMMI meaning, the ordinary dictionary meaning applies. This is a selected subset of the model's glossary.

Term	Meaning
5-Whys	A technique for finding an issue's underlying causes by repeatedly asking 'Why?' until the cause is identified.
Acceptance criteria	The criteria a solution must satisfy to be accepted by customers.
Acquirer	The stakeholder who obtains a solution from a supplier.
Affected stakeholders	People impacted by a process, activity, work product, or decision.
Allocated requirement	A requirement resulting from levying all or part of a higher-level requirement on a lower-level design component.
Appraisal	An examination of one or more processes by a trained team using a reference model as the basis for determining, at minimum, strengths and weaknesses.
Architecture	The set of structures needed to establish a solution, their components, relationships, and properties.
Base measure	A measure that is functionally independent of other measures, defined in terms of an attribute and a method for quantifying it.
Baseline	A set of specifications or work products formally reviewed and agreed on, serving as the basis for further work, changeable only through change control.
Benchmark Model View	A logical grouping of predefined CMMI components used to define appraisal model view scope (defined in the model's Appendix B).
Bidirectional traceability	An association allowing tracing in either direction between logical entities, e.g. requirements to design to code to test.
Business objectives	The objectives the organization sets to achieve its strategy; all CMMI improvement and measurement objectives should trace to them.
Capability Area (CA)	A group of related Practice Areas that together improve performance in a set of skills and activities; a type of view.
Capability level	A rating of performance and process improvement achievement in individual PAs (maximum Capability Level 3; GOV and II must be in scope).
Category	A logical group of related Capability Areas addressing common business problems (Doing, Managing, Enabling, Improving).
Context specific information	Informative material relevant to a particular industry, methodology, or discipline; identified by a context tag.
Core Practice Area	A PA that applies regardless of domain; the 16 core PAs are included in every maturity level view.
Domain	A functionally similar grouping of PAs aligned to an organization's primary capability (Data, Development, People, Safety, Security, Services, Suppliers, Virtual).
Explanatory information	Informative material that applies to all contexts and helps users understand intent and business value; includes example activities and work products.
Habit	A tendency or practice, especially one that is hard to give up.
High maturity	The application of statistical and other quantitative techniques to selected processes (Practice Group Levels 4 and 5).

Term	Meaning
Intent	Required PA information explaining what results and accomplishments are expected from the PA.
Maturity level	A rating of performance and process improvement achievement across a predefined set of PAs (core PAs plus one or more domains). Levels build on each other and cannot be skipped.
Practice Area (PA)	A set of practices that collectively describe the critical activities needed to achieve a defined intent and value.
Practice group	The organizing structure for practices within a PA; practice group levels 1-5 provide an evolutionary path of increasing capability.
Process asset library	The repository holding the organization's processes and process assets, made available for use.
Process performance baseline (PPB)	A documented characterization of actual process performance, used to compare expected against actual results.
Process performance model (PPM)	A description of relationships among measurable process attributes and outcomes, used to predict results.
QPPO	Quality and Process Performance Objectives - quantitative objectives traceable to business and performance objectives; the starting point of high maturity.
Persistence	Firm or stubborn continuance in a course of action despite difficulty or opposition.
Solution	The product, service, or other output delivered to a customer or end user.
Special cause of variation	Variation from an unusual, assignable event; understood and managed at Practice Group Level 4.
Common cause of variation	Variation inherent in the process itself; understood and improved at Practice Group Level 5.
Tailoring	Adapting the organization's standard processes and assets to the unique characteristics of a project or work effort, using recorded criteria and guidelines.
Value	Required PA information stating the business value achievable by adopting the practices in the PA.
View	A window into the model that lets an organization focus on what matters to it; may be predefined (domain, capability area, maturity level) or custom.

12. Abbreviations

Abbr.	Meaning	Abbr.	Meaning
CA	Capability Area	**PA**	Practice Area
CAR	Causal Analysis and Resolution	**PAD**	Process Asset Development
CM	Configuration Management	**PCM**	Process Management
CMMI	Capability Maturity Model Integration	**PG**	Practice Group
CONT	Continuity	**PI**	Product Integration
DAR	Decision Analysis and Resolution	**PLAN**	Planning
DM	Data Management	**PPB**	Process Performance Baseline
DQ	Data Quality	**PPM**	Process Performance Model
ESAF	Enabling Safety	**PQA**	Process Quality Assurance
ESEC	Enabling Security	**PR**	Peer Reviews
EST	Estimating	**QPPO**	Quality and Process Performance Objective
EVW	Enabling Virtual Work	**RDM**	Requirements Development and Management
GOV	Governance	**RSK**	Risk and Opportunity Management
II	Implementation Infrastructure	**SAM**	Supplier Agreement Management
IRP	Incident Resolution and Prevention	**SDM**	Service Delivery Management
MC	Monitor and Control	**SHP**	Sustaining Habit and Persistence
MDD	Method Definition Document	**STSM**	Strategic Service Management
ML	Maturity Level	**TS**	Technical Solution
MPM	Managing Performance and Measurement	**VV**	Verification and Validation
MST	Managing Security Threats and Vulnerabilities	**WE**	Workforce Empowerment
OT	Organizational Training		

Document control. CMMI Model V3.0 was released 6 April 2023; change information is published in the CMMI model release notes on the CMMI Institute site. Adoption resources — the CMMI Adoption Guidance, Tech Talks, the CMMI Performance Report Summary, and Published Appraisal Results (PARS) — are available through the CMMI Resource Center.